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CODER: Knowledge-infused cross-lingual medical term embedding for term normalization

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ABSTRACT

Objective: This paper aims to propose knowledge-aware embedding, a critical tool for medical term normalization.
Methods: We develop CODER (**C**ross-lingual kn**o**wledge-infused me**d**ical t**er**m embedding) via contrastive learning based on a medical knowledge graph (KG) named the Unified Medical Language System, and similarities are calculated utilizing both terms and relation triplets from the KG. Training with relations injects medical knowledge into embeddings and can potentially improve their performance as machine learning features.
Results: We evaluate CODER based on zero-shot term normalization, semantic similarity, and relation classification benchmarks, and the results show that CODER outperforms various state-of-the-art biomedical word embeddings, concept embeddings, and contextual embeddings.
Conclusion: CODER embeddings excellently reflect semantic similarity and relatedness of medical concepts. One can use CODER for embedding-based medical term normalization or to provide features for machine learning. Similar to other pretrained language models, CODER can also be fine-tuned for specific tasks. Codes and models are available at https://github.com/GanjinZero/CODER.

1. Introduction

Electronic medical records (EMRs) contain abundant medical information that can be used in various medical tasks, such as clinical decision support [1], automated phenotyping [2], and medical knowledge mining [3]. However, analyzing the free text in EMRs remains challenging because the same medical concept can appear with various nonstandard names, abbreviations, and misspellings, which all need to be normalized to the corresponding standard terms or the concept IDs of an existing terminology system, such as the Unified Medical Language System (UMLS) [4] or SNOMED Clinical Terms (SNOMED-CT) [5]. Normalization is also important for structured EMR data. Different medical institutes use different coding systems for laboratory examinations, procedures, and medications. Thus, when conducting multi-institutional studies, normalization can facilitate mapping database fields to the same standard for joint analysis.

Medical term normalization can be regarded as either a classification or a ranking problem. Neural-network-based techniques encode terms into hidden representations based on recurrent neural networks or pretrained language models (PLMs). A softmax layer on the hidden representations [6–9] is used for classification. Classification methods achieve state-of-the-art scores in benchmark tests but are limited to a small set of target concepts. In practice, the terminology systems in use, such as the UMLS, encompass millions of concepts. Normalizing medical terms to these systems remains challenging in classification due to the enormous number of classes and the inflexibility of adding new classes. Another approach is to rank candidate concepts by similarity to a term [10–12]. This method differs from the previous one in that training is based on a binary classifier, and terms and the corresponding concept names (target terms) form positive samples; terms and non-corresponding concept names form negative samples. The scalar output of the classifier is then used as a measure of similarity to rank the candidate concepts for normalization. Similar to the classification approach, these models are trained based on labeled datasets with limited target concepts.

To achieve term normalization for very large-scale and flexible

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concept sets, we propose CODER based on PLM with KG contrastive learning. CODER generates embeddings for terms and ranks the candidate concepts by the cosine similarity between the embeddings of the candidate target terms and those of the input term, thereby avoiding the training of a classification model. The existing medical embeddings can be classified into word embeddings [13–15], concept embeddings [16–21], and contextual embeddings [21–28]. Word and concept embeddings have displayed good base performances in evaluating similarity but face the out-of-vocabulary (OOV) problem and cannot handle misspellings that are commonly present in clinical text. PLM-based contextual embeddings can alleviate OOV problems by using subword tokenization, but they are outperformed by word and concept embeddings in evaluating similarity if not fine-tuned [29].

Contrastive learning is a popular technique in computer vision (CV) [30,31] and can learn representations with a very large number of classes. In each training step, instead of learning to map samples to their labels, contrastive learning aims to learn sample representations to distinguish samples with the same label (positive pairs) from those with different labels (negative pairs). For term normalization, terms related to the same and different concepts naturally serve as positive and negative pairs in contrastive learning, as has been explored with SapBERT [32] using the UMLS Metathesaurus. The UMLS Metathesaurus contains synonymous names for medical concepts integrated from different lexicon resources in multiple languages. However, as a KG, the UMLS contains more knowledge than terms alone, specifically, relational triplets. These relational knowledge can serve as useful information for normalization. For example, when normalizing *poisoned by eating pufferfish,* we may want *food poisoning* to rank among the top results because pufferfish is a kind of food. As another example, it would be more meaningful to embed *rheumatoid arthritis* closer to *osteoarthritis* than to *rheumatoid pleuritis* because both *rheumatoid arthritis* and *osteoarthritis* are subtypes of arthritis that may affect the same body parts and share treatments. The reasoning in these examples illustrates that relational knowledge is essential for achieving meaningful medical embeddings. Therefore, we propose dual contrastive learning based on both the terms and relation triplets in KGs. The term-relation-term similarity is measured between a term-relation (h, r) and a term (t). Positive and negative term-relation-term pairs were determined based on KGs. Online hard pair mining is applied to find informative pairs that are often used in person reidentification (ReID) [33]. ReID and medical term normalization are similar tasks since they both involve many categories and sparse samples for each category.

To support cross-lingual term normalization, we use mBERT [34], which encodes texts in different languages, to one unified space. Previous methods were usually translation based and relied on biomedical parallel corpora [35–37]. UMLS naturally contains massive cross-lingual medical synonyms that have not been fully utilized.

The contributions of this paper are as follows:

1. We propose a KG-based contrastive learning model for term normalization that uses both synonyms and relations from UMLS to direct the generation of medical term embeddings. The learning strategy can also be applied to other KGs.
2. We perform evaluations of CODER with notable existing medical embeddings. CODER achieves state-of-the-art results in zero-shot term normalization, medical concept similarity measurement, and concept relation classification tasks. CODER embeddings can be directly used for embedding-based term normalization or as features for machine learning, and CODER can be fine-tuned like other PLMs.
3. CODER is a cross-lingual medical term representation with support for English, Czech, French, German, Italian, Japanese, Portuguese, Russian, Spanish, Dutch, and Chinese.

2. Related work

In this section, we briefly introduce the medical term normalization task, existing medical embeddings, and contrastive learning.

2.1. Medical term normalization

Classification methods commonly generate hidden representations of terms and use a softmax layer to classify terms into concepts. Terms are encoded by CNNs, RNNs, or PLMs [6,7,9]. An attention mechanism is introduced to capture important words or characters for classification [7,8]. These classification methods only consider target terms as a category and do not utilize semantic information from target terms.

Ranking methods rank the similarity between an input term and candidate target terms by training terms as positive and negative pairs. DNorm [10] learns to rank target terms by calculating the similarities between TF-IDF vectors. Li et al. [11] encoded input and target terms with a CNN and used a pairwise approach for ranking. NSEEN [38] trained a Siamese LSTM network and used hard negative samplings to find informative pairs. BNE [39] encoded terms, concepts, and contexts separately, and model training is based on term-term, term-concept, and term-context similarities. Pattisapu et al. [40] calculated graph embeddings for target terms and ranked them based on the cosine distance. BIOSYN [12] uses both TF-IDF and BioBERT [25] to represent terms. Moreover, synonym marginalization is used to maximize the similarity between synonyms.

Non-English medical term normalization has attracted the interest of researchers. In Ref. [8], the Chinese Medical Concept Normalization dataset was used. Translation-based methods have been used for French, Spanish, Dutch, and German [35–37].

2.2. Medical embeddings

Word embeddings are typically trained with the word2vec model [41] using medical records or papers [13–15].

Concept embeddings map medical concepts to embedding vectors and cannot perform term normalization intrinsically. Sequences of medical concepts identified from EMRs or biomedical papers are considered as sentences to train embeddings with word2vec [16–19]. Cui2vec [20] factorizes the CUI-CUI pointwise mutual information matrix to obtain embeddings. Ref. [21] proposed conceptual-contextual embedding trained on both biomedical corpora and UMLS relations.

Contextual embedding methods can generate different embeddings according to context. The popularity of contextual embeddings stems from large-scale PLMs such as BERT [34] and ELMo [42]. Several medical contextual embeddings have been trained on different medical corpora in the BERT framework [22–27]. Jin et al. [28] proposed BioELMo training based on PubMed. SapBERT [32] conducts self-alignment pretraining with UMLS synonyms for term embedding. Contextual embeddings have excellent potential for use in various kinds of downstream NLP tasks but typically require supervised fine-tuning. We found that most existing contextual embeddings perform worse in zero-shot term normalization than word embedding since the [CLS] token of masked language models may not directly reflect semantic similarity [29].

2.3. Contrastive learning

Contrastive learning has been successfully used to learn representations by comparing positive and negative pairs, especially in the field of CV [43]. SimCLR [31], MoCo [30] and SwAV [44] have achieved comparable results as supervised methods for the ImageNet dataset. Data augmentation is required for self-supervised contrastive learning to generate positive pairs. Crop, Gaussian blur, and color distortion [31] can be used in the CV domain. For NLP, DeCLUTR [45] collects positive sentence pairs from the same document. Back translation is applied in CERT [46] to create similar sentence pairs. SapBERT [32] utilizes synonyms from the UMLS as positive pairs. Compared to SapBERT, our work fully utilizes the UMLS as a KG and performs training based on both synonyms and relation triplets. Loss functions are another essential

component for contrastive learning to maximize the similarity between positive pairs and minimize the similarity between negative pairs. Triplet loss [47] minimizes the relative distance between one positive and one negative pair. InfoNCE [48] considers all other samples in a minibatch as negative pairs.

3. Method

In this section, we introduce the model architecture of CODER. We first define the embedding-based term normalization task and how we embed terms. Then, we introduce KG-based contrastive learning for term normalization.

3.1. Embedding-based term normalization

Let $\mathcal{D} = \{n_i\}_{i=1}^{\|\mathcal{D}\|}$ be a concept dictionary, where n_i is a concept. Each concept n_i is associated with several terms $\left\{s_i^j\right\}_{j=1}^{c_i}$ as synonyms. We define the embedding assigned to term s_i^j as $\boldsymbol{e}_i^j \in \mathbb{R}^l$. For an input string s with embedding $\boldsymbol{e}$, the goal is to predict the corresponding concept n:

$$\widehat{n} = n_{\mathrm{argmax}_i\left(\cos\left(\boldsymbol{e}_i^j, \boldsymbol{e}\right)\right)},$$

where cos is the cosine similarity measure in Euclidian space.

3.2. Term representation

CODER takes an input term s and outputs an embedding vector $\boldsymbol{e}$. The input term s is tokenized to subwords $[\mathrm{CLS}], s_0, s_1, \cdots, s_k, [\mathrm{SEP}]$ with two special tokens, [CLS] (added to the head of a sentence) and [SEP] (added to the tail of a sentence). A pretrained language model PLM encodes s to a series of hidden states $\boldsymbol{h}_i$:

$$\boldsymbol{h}_{[\mathrm{CLS}]}, \boldsymbol{h}_0, \cdots, \boldsymbol{h}_k, \boldsymbol{h}_{[\mathrm{SEP}]} = \mathrm{PLM}([\mathrm{CLS}], s_0, \cdots, s_k, [\mathrm{SEP}]).$$

The structure of PLM is the same as a BERT model which is composed of a 12-layer transformer encoder and initialized by PubMedBERT [26] or mBERT [34]. The [CLS] token has an attention-based weight to all the tokens from the layer below, and we use it to represent the term s:

$$\boldsymbol{e} = \boldsymbol{h}_{[\mathrm{CLS}]}.$$

3.3. The contrastive learning framework for KGs with synonyms

CODER learns term representations by maximizing the similarities between positive term-term pairs and term-relation-term pairs from a KG. The KG records relation triplets (h, r, t), where the head concept h and the tail concept t are from concept dictionary $\mathcal{D}$ and $r \in \mathcal{R}$ is the relation between them. In each training step, we sample a batch of relation triplets $(h_i, r_i, t_i)_{i=1}^k$ from the KG. For convenience, we denote $n_i = h_i, n_{k+i} = t_i$, where$\forall 1 \le i \le k$. We sample term s_{I} for each n_i from $\left\{s_i^j\right\}_{j=1}^{c_i}$ in $\mathcal{D}$. CODER embeds terms $\{s_i\}_{i=1}^{2k}$ into $\{e_i\}_{i=1}^{2k}$.

We define the term-term pair label τ_{ij} in a minibatch based on the following relation set:

$$\tau_{ij} = \begin{cases} 1, & n_i = n_j \\ 0, & n_i \neq n_j. \end{cases}$$

The similarity between two terms in a minibatch is defined as:

$$S_{ij} = \mathrm{cosine}\left(\boldsymbol{e}_i, \boldsymbol{e}_j\right).$$

Various pair-based loss functions, such as InfoNCE [48], have been proposed for learning representations in contrastive learning. We adopt Multi-Similarity loss (MS-loss) [49] to learn embeddings (InfoNCE loss and triplet loss have been tested in our preliminary experiments and showed worse performances); this metric incorporates hard pair mining in its formulation to accelerate convergence. Other pair-based losses can also be used to establish a loss function, and we leave comparative analyses of loss functions for future work. MS-loss can be used to mine hard pairs according to positive relative similarity. For an anchor i, the following relations are defined:

$$\mathcal{N}_i = \left\{j | \tau_{ij} = 0, \quad S_{ij} > \min_{\tau_{ik}=1} S_{ik} - \epsilon\right\},$$

$$\mathcal{P}_i = \left\{j | \tau_{ij} = 1, \quad S_{ij} < \max_{\tau_{ik}=0} S_{ik} + \epsilon\right\},$$

where ϵ is a margin. The MS-loss of a mini-batch is:

$$\mathcal{L}_{MS} = \frac{1}{2k}\sum_{i=1}^{2k}\left(\frac{\log\left(1 + \sum_{j\in\mathcal{P}_i}\exp\left(-\alpha\left(S_{ij} - \lambda\right)\right)\right)}{\alpha} + \frac{\log\left(1 + \sum_{j\in\mathcal{N}_i}\exp\left(\beta\left(S_{ij} - \lambda\right)\right)\right)}{\beta}\right),$$

where $\alpha, \beta,$ and λ are hyperparameters. $\mathcal{P}_i$ will vanish if $\tau_{\mathrm{ij}} = 0$, with $\forall \mathrm{j} \neq \mathrm{i}$, and $\mathcal{N}_i$ will vanish if $\tau_{ij} = 1$, with $\forall j \neq i$.

To leverage relational knowledge from a KG into term representations, CODER also learns KG embeddings based on the approach used in semantic matching methods such as DistMult [50] and ANALOGY [51] to approximate $\boldsymbol{M}_r^\top h \approx t$. This is the main difference between CODER and SapBERT. Considering two triplets (h_0, r, t_0) and (h_1, r, t_1) with the same relation type, if h_0 and h_1 are semantically similar (e.g., similar diseases), t_0 and t_1 may also be semantically similar (e.g., similar drugs). $\boldsymbol{M}_r^\top h \approx t$ will guarantee relation-based similarity inference in the above example.

(h_i, r_i) can be viewed as a relation-based data augmentation of t_i. We can define the term-relation-term similarity between relation $(\mathrm{s_I}, \mathrm{r_I})$, with $\forall 1 \le \mathrm{i} \le \mathrm{k}$, and term $\mathrm{s_j}$, with $\forall \mathrm{k} + 1 \le \mathrm{j} \le 2\mathrm{k}$, in a minibatch as:

$$\mathrm{S}_{\mathrm{ij}}^{\mathrm{rel}} = \cos\left(\boldsymbol{M}_{\mathrm{r_i}}^\top \boldsymbol{e}_i, \boldsymbol{e}_j\right) = \frac{\boldsymbol{e}_i^\top \boldsymbol{M}_{r_i} \boldsymbol{e}_j}{\|\boldsymbol{M}_{\mathrm{r_i}}^\top \boldsymbol{e}_i\| \|\boldsymbol{e}_j\|},$$

where $\boldsymbol{M}_{r_i} \in \mathbb{R}^{l\times l}$ and $\|\cdot\|$ is the L2-norm. Instead of using $\boldsymbol{e}_i^\top \boldsymbol{M}_{r_i} \boldsymbol{e}_j$ as the score function, CODER uses cos to bound S_{ij}^{rel}. When $\tau_{ij} = 1$, S_{ij} can be considered a special case of S_{ij}^{rel} with $\boldsymbol{M}_{r_i} = \boldsymbol{I}$. The original DistMult samples corrupted triplets (h', r, t') as negative training sample of (h, r, t), but CODER utilizes $\left(h_i, r_i, t_{j-k}\right)$ as negative term-relation-term pairs in a minibatch (recall that $\mathrm{k}+1 \le \mathrm{j} \le 2\mathrm{k}$ and that the index for tail entities ranges from 1 to k).

The term-relation-term pair label $\tau_{\mathrm{ij}}^{\mathrm{rel}}$ is defined by:

$$\tau_{\mathrm{ij}}^{\mathrm{rel}} = \begin{cases} 1, & \left(h_i, r_i, t_{j-k}\right) \in KG, \\ 0, & \left(h_i, r_i, t_{j-k}\right) \notin KG. \end{cases}$$

It is time-consuming to determine if a relation triplet is included in the relevant KG. In practice, we calculate:

$$\tau_{ij}^{rel} = \begin{cases} 1, & t_i = t_{j-k}, \\ 0, & t_i \neq t_{j-k}. \end{cases}$$

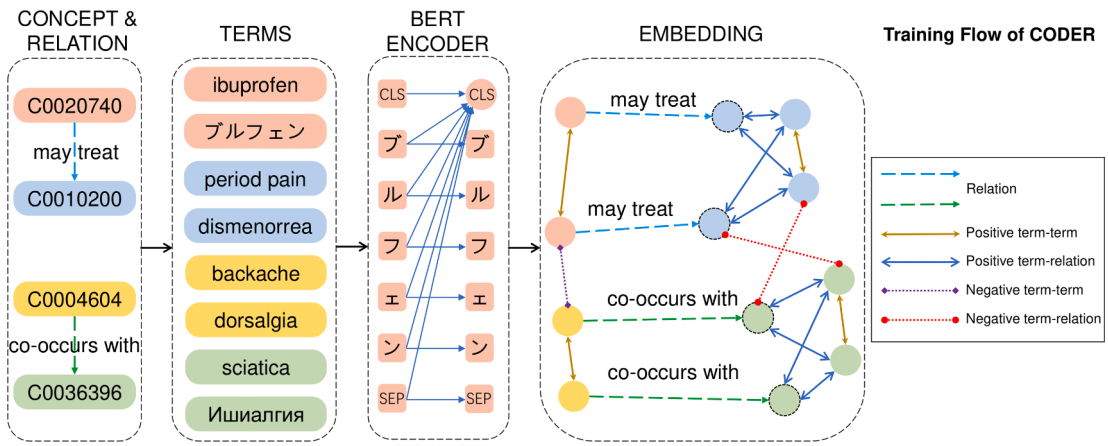

Fig. 1. The overview of CODER. CODER encodes terms potentially in different languages in the embedding space. Term-term similarities and term-relation-term similarities are calculated to train CODER.

Table 1
acc@k for different embeddings in the Cadec and PsyTar datasets.

Embedding/Method	Cadec		PsyTar	
	acc@1	acc@3	acc@1	acc@3
GoogleNews-vectors	39.15	52.41	29.75	47.52
Wikipedia-PubMed-PMC	35.04	47.14	23.97	36.02
BioNLP-2016-win-2	35.47	48.30	25.29	37.70
BioNLP-2016-win-30	38.78	51.64	28.27	43.26
SciSpacy	48.69	60.26	46.04	62.39
QuickUMLS	44.18	56.51	43.31	58.93
BERT-base-cased	10.63	14.97	10.50	14.88
BioBERT v1.1	11.68	15.71	5.33	7.96
ClinicalBERT (Huang)	9.65	13.37	7.37	10.61
ClinicalBERT	9.99	12.82	7.07	9.56
SciBERT	15.85	21.85	13.18	18.68
BlueBERT	10.44	14.06	9.07	11.95
PubMedBERT	15.17	21.33	14.05	19.43
SapBERT	57.56	73.34	54.15	71.71
$\text{CODER}_{\text{ENG}}$	**59.01**	**76.19**	**55.62**	**72.91**
$\text{CODER}_{\text{ALL}}$	56.02	72.11	43.86	60.20

With S_{ij}^{rel} and τ_{ij}^{rel} defined, MS-loss can also be used to learn knowledge embeddings. CODER mines hard term-relation-term pairs for anchor (s_i, r_i) based on:

$$\mathcal{N}_i^{rel} = \left\{ j | \tau_{ij}^{rel} = 0,\ \ S_{ij}^{rel} > \min_{\tau_{ik}^{rel}=1} S_{ik}^{rel} - \epsilon^{rel} \right\},$$

$$\mathcal{P}_i^{rel} = \left\{ j | \tau_{ij}^{rel} = 1,\ \ S_{ij}^{rel} < \min_{\tau_{ik}^{rel}=0} S_{ik}^{rel} + \epsilon^{rel} \right\},$$

where ϵ^{rel} is a margin. The MS-loss of term-relation-term pairs is:

$$\mathcal{L}_{MS}^{rel} = \frac{1}{k} \sum_{i=1}^{k} \left(\frac{\log\left(1 + \sum_{j \in \mathcal{P}_i^{rel}} \exp\left(-\alpha^{rel}\left(S_{ij}^{rel} - \lambda^{rel}\right)\right)\right)}{\alpha^{\text{rel}}} + \frac{\log\left(1 + \sum_{j \in \mathcal{N}_i^{rel}} \exp\left(\beta^{rel}\left(S_{ij}^{rel} - \lambda^{rel}\right)\right)\right)}{\beta^{rel}} \right),$$

where $\alpha^{rel}, \beta^{rel}$, and λ^{rel} are hyperparameters. To avoid the degeneration of $\mathcal{P}_i, \mathcal{N}_i, \mathcal{P}_i^{rel}$, and $\mathcal{N}_i^{rel}$, we guarantee that for each triplet (h_i, r_i, t_i), at least m different indexes $\{\mathrm{i_o}\}_{\mathrm{o}=1}^{\mathrm{m}}$ exist, such that $(\mathrm{h_I}, \mathrm{r_I}, \mathrm{t_i}) = (\mathrm{h_{i_o}}, \mathrm{r_{i_o}}, \mathrm{t_{i_o}})$, where $2 \le \mathrm{m} \le \sqrt{\mathrm{k}}$.

The total loss is formulated as:

$$\mathcal{L} = \mathcal{L}_{\text{MS}} + \mu \mathcal{L}_{MS}^{rel}$$

The parameters of CODER are updated to minimize $\mathcal{L}$. The overall model structure of CODER is illustrated in Fig. 1.

4. Experiments

In this section, we first introduce the UMLS and the training detail of CODER. Then, we introduce the compared embeddings and the benchmarks for medical term normalization tasks, a medical similarity measurement, and relation classification based on certain concepts.

4.1. Training settings of CODER

We use the UMLS 2020AA release, which contains 4.27 M concepts, 15.48 M terms, and 87.89 M relations with 127 semantic types, 14 relation types, and 923 relationship attributes to build the training dataset. We concatenate the names of the relation types and the attributes as the relation labels of triplets. We initialized the parameters of $\text{CODER}_{\text{ENG}}$ from PubMedBERT and trained it with 100 K training steps. We also trained the cross-lingual version $\text{CODER}_{\text{ALL}}$ initialized by mBERT with 1 M training steps. We use [CLS] representation in the final layer as term representation. A batch size of $\mathrm{k} = 128$ relation triplets and 8 gradient accumulation steps were used for training. In each minibatch, we set the count of repeat triplets to $\mathrm{m} = 8$. The maximal sequence length was 32 since the terms are short. We used AdamW [52] as the optimizer with a linear warm-up period over the first 10,000 steps to reach a peak learning rate of 2e-5 that linearly decayed to zero. The parameters $\alpha = \alpha^{\text{rel}}, \beta = \beta^{\text{rel}}, \lambda = \lambda^{\text{rel}}$, and $\epsilon = \epsilon^{\text{rel}}$ in the MS-loss function were set to 2, 50, 0.5, and 0.1. The weight μ was set to 1 to balance term-term loss and term-relation-term loss.

4.2. Compared baseline embeddings

We compare CODER with the following medical embeddings. Details of the compared embeddings are listed in Appendix A.1.

Word embedding GoogleNews vector [41] is a baseline of word

Table 2
F1-scores for the MANTRA GSC based on translation and embedding-based methods.

Method	Medline				EMEA				Patent	
	SPA	FRE	DUT	GER	SPA	FRE	DUT	GER	FRE	GER
BTM	69.1	67.4	61.4	66.3	–	–	–	–	–	–
GB	68.7	**68.6**	**64.8**	67.9	–	–	–	–	–	–
mBERT	13.0	12.2	12.3	16.3	16.9	25.5	16.8	16.1	23.2	22.8
$\text{CODER}_{\text{ALL}}$	**70.1**	58.6	58.6	**69.0**	**68.1**	**62.9**	**61.7**	**65.3**	**70.8**	**69.0**

Table 3
MCSM scores for different embeddings.

Embedding	PS	DS	NP	CD	FD	IP
cui2vec	4.2	7.2	7.6	4.3	7.2	8.4
DeVine-etal-200	7.3	6.9	6.3	1.2	4.6	2.5
claims-codes-hs-300	9.4	7.7	7.5	–	3.4	8.6
GoogleNews-vectors	9.1	9.3	9.6	10.0	8.5	10.3
Wikipedia-PubMed-PMC	8.6	9.4	10.0	9.8	8.9	10.3
BioNLP-2016-win-2	9.1	9.4	10.0	10.2	8.9	**10.4**
BioNLP-2016-win-30	8.3	9.3	10.0	9.6	8.8	10.3
BERT-base-cased	9.2	8.4	8.2	10.0	7.5	9.2
BioBERT v1.1	9.2	9.2	9.7	10.2	7.9	9.8
ClinicalBERT (Huang)	7.2	6.5	6.2	8.6	5.6	8.6
ClinicalBERT	9.1	8.9	9.2	9.5	7.4	9.5
SciBERT	9.7	9.2	9.3	10.3	8.8	10.0
BlueBERT	8.2	8.2	9.1	9.6	7.4	9.3
PubMedBERT	8.9	9.1	9.7	10.2	9.0	9.9
SapBERT	9.8	**9.6**	9.3	9.7	9.3	10.1
$\text{CODER}_{\text{ENG}}$	**10.4**	**9.6**	9.7	**10.7**	9.6	10.1
$\text{CODER}_{\text{ALL}}$	10.2	9.5	**10.1**	10.7	**9.8**	10.3

Table 4
Testing accuracy of DDBRC for training 50 epochs.

Embedding	Features	Fine-tuned
cui2vec	88.95	88.92
DeVine-etal-200	80.42	80.44
claims-codes-hs-300	72.02	71.99
GoogleNews-vectors	90.96	90.85
Wikipedia-PubMed-PMC	91.67	91.16
BioNLP-2016-win-2	91.94	92.00
BioNLP-2016-win-30	92.00	91.46
BERT-base-cased	78.22	98.74
BioBERT v1.1	83.47	98.73
ClinicalBERT (Huang)	75.82	98.77
ClinicalBERT	76.12	98.76
SciBERT	77.79	98.80
BlueBERT	79.70	98.78
PubMedBERT	75.93	98.81
SapBERT	91.80	98.79
$\text{CODER}_{\text{ENG}}$	94.45	**98.90**
$\text{CODER}_{\text{ALL}}$	**95.82**	98.81

embedding. Wiki-PubMed-PMC [13], BioNLP-2016-Win-2 [14], and BioNLP-2016-Win-30 [14] are medical word embeddings with different training settings. To represent a term, we used the average of the embeddings of its words.

Concept embedding Cui2vec [20], DeVine-etal-200 [16], and claims-codes-hs-300 [17] map different sets of medical concepts to vectors. Cui2vec contains medical concepts from SNOMED-CT. DeVine-etal-200 includes medical concepts that appeared in MedTrack and OHSUMED. Additional, claims-codes-hs-300 contains ICD-9 codes, LOINC laboratory codes, and NDC medication codes.

Contextual embedding BERT-base-cased and mBERT [34] are the baselines of contextual embedding. BioBERT [25] is the most well-known biomedical language model. Two clinical language models that have the same name, ClinicalBERT [22,23], exist, and we distinguish them based on the author's name. We compare SciBERT [24], BlueBERT [27] and PubMedBERT [26], which are biomedical-specific language models. We also compare SapBERT [32], which is a BERT-based medical term representation model trained using contrastive learning with UMLS terms. We use the [CLS] representation as the term embedding, as described in Section 3.2.

4.3. Medical term normalization tasks

We evaluate term normalization performance based on three datasets in different languages. Term normalization tasks are evaluated in a zero-shot setting, i.e., no training datasets are provided.

4.3.1. Cadec and PsyTar

Cadec [53] and PsyTar [54] are two English medical term normalization datasets. Cadec contains 6754 terms and 1029 standard terms as normalization targets. PsyTar contains 6556 medical terms, which are mapped to 618 standard terms. We use the data division schemes from Refs. [7,9] and the top-k accuracy as the metric. We report the average scores for different test sets. Besides comparing with word embeddings and contextual embeddings, we also compare CODER with QuickUMLS [55] and SciSpacy [56], which utilizes n-gram features to calculate term similarities.

Table 5
Ablation results for different training settings in CODER.

Setting	Cadec		PsyTar	
	acc@1	acc@3	acc@1	acc@3
$\text{CODER}_{\text{ENG}}$	59.01	76.19	55.61	72.91
$-S_{ij}^{rel}$	54.91	70.84	49.92	67.28
$-S_{ij}^{rel} + S_{ij}^{rel.T}$	57.35	74.46	54.65	71.83

The results for Cadec and PsyTar are shown in Table 1. For both datasets, CODER achieved superior performance over other baseline embedding methods and n-gram based methods. To notice, although the output terms (i.e. target concepts) in Cadec and PsyTAR are from the UMLS, the input terms (i.e. terms need to normalize) are not. CODER has not seen these input terms in the training dataset before. $\text{CODER}_{\text{ENG}}$ based on KG contrastive learning outperformed SapBERT, which only uses synonyms, by 1.45, 2.85, 1.47, and 1.20 in terms of Cadec acc@1, acc@3, PsyTar acc@1, and acc@3, respectively. $\text{CODER}_{\text{ALL}}$ is trained in multiple languages, which may have caused $\text{CODER}_{\text{ALL}}$ to perform slightly worse than $\text{CODER}_{\text{ENG}}$. Additionally, we observed the following interesting phenomena:

(1) Contextual embeddings pretrained with masked language models failed to achieve embedding-based normalization.
(2) Medical word embeddings do not improve the word embedding baseline.
(3) CODER has better performance than other PLMs, showing that pretraining using UMLS synonyms and relational knowledge (compared to masked language model task) not only help the model learn how to normalize terms in the UMLS, but also generalize to terms that appear on social media.

4.3.2. The MANTRA GSC

The MANTRA GSC [57] contains French, German, Spanish, and

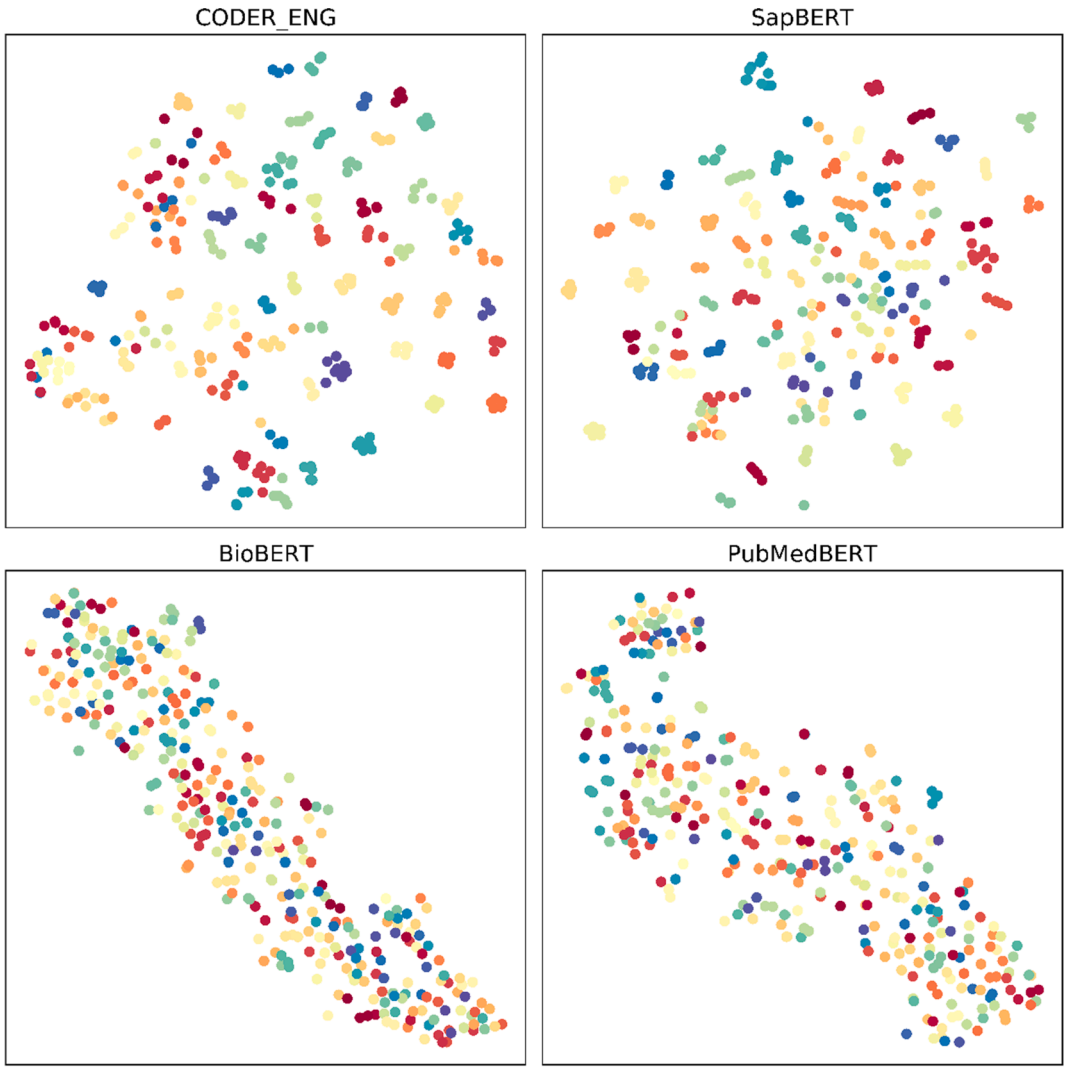

Fig. 2. The t-SNE visualization for different contextual embeddings with term normalization.

Dutch corpora and includes medical terms mapped to a subset of the UMLS from three different corpora (MEDLINE, EMEA, and Patent). To our knowledge, only some of these non-English corpora have been evaluated before. We compare $\text{CODER}_{\text{ALL}}$ with two translation-based methods: the biomedical translation model (BTM) [36] and a combination of Google Translator and Bing Translator (GB) [36]. We also compare with mBERT as a multi-lingual contextual embedding baseline.

$\text{CODER}_{\text{ALL}}$ and mBERT search for standard terms in SNOMED-CT, MeSH, and MedDRA of UMLS2020AA, which includes 801,038 concepts, and return one CUI per input term according to the cosine similarity. We use the F1-score as the metric since one term can map to multiple CUIs in the MANTRA GSC.

Table 2 summarizes the MANTRA results. $\text{CODER}_{\text{ALL}}$ benefits from UMLS multilanguage synonyms and exhibits obvious improvements over mBERT. The BTM is trained using an additional dictionary, and the GB is trained with a massive parallel corpus and achieves better performance than $\text{CODER}_{\text{ALL}}$ for the French and Dutch MEDLINE datasets. $\text{CODER}_{\text{ALL}}$ achieves state-of-the-art performance for the Spanish and German MEDLINE, EMEA, and Patent corpora.

4.4. MCSM

Choi et al. [17] introduced the medical conceptual similarity measure (MCSM) for evaluating medical embeddings. The MCSM is defined by:

$$MCSM(V,T,k) = \frac{1}{|V(T)|}\sum_{v\in V(T)}\sum_{i=1}^{k}\frac{1_T(v(i))}{\log_2(i+1)},$$

where V is a set of medical concepts, T is a semantic type, $|V(T)|$ is the number of medical concepts associated with semantic type T, k is the neighborhood size, $v(i)$ is the ith closest neighbor, and $1_T(\cdot)$ is an indicator function for semantic type T. We reimplement the evaluation code of the MCSM to support word and contextual embeddings.

We use SNOMED-CT, which contains 157,726 concepts, as V. Based on the settings in Ref. [17], pharmacologic substance (PS), disease or syndrome (DS), neoplastic process (NP), clinical drug (CD), finding (FD), and injury or poisoning (IP) are the evaluated semantic types T, and k is set to 40.

The MCSM results are reported in Table 3. Concept embeddings suffer from OOV issues, cannot cover all SNOMED-CT concepts and perform poorly in conjunction with the MCSM. Although CODER is not designed for semantic classification or clustering, it still learns semantic information via relations from the UMLS and achieves state-of-the-art performance, and CODER is only improved by word embeddings by a small margin for IP. CODER shows consistent improvement in each semantic type compared to SapBERT by utilizing relational knowledge.

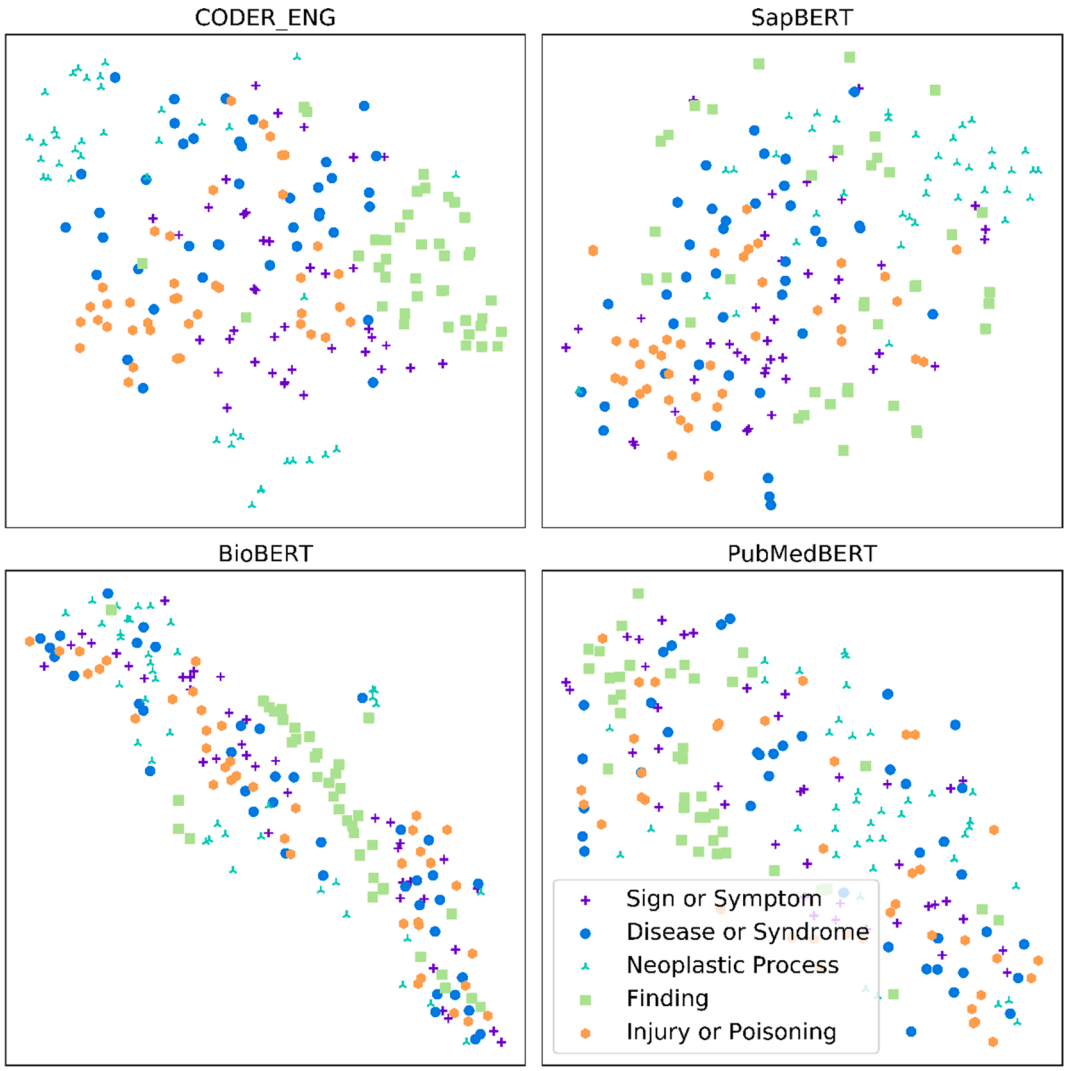

Fig. 3. The t-SNE visualization of different embeddings for concepts associated with different semantic types.

Table 6
The closest term to **C0004604** *dorsalgia* in different languages.

Language	Term	Target
Spanish	dolor de espalda	dorsalgia
Hungarian	Hátfájás	dorsalgia
Japanese	背部痛	dorsalgia
Japanese	はいぶつう	puberty
Czech	bolesti zad	dorsalgia
Germany	Rückenschmerzen	dorsalgia
Italian	Dolore dorsale	dorsalgia
Korean	배통	benign neoplasm of ureter
Finnish	Selkäkipu	dorsalgia
French	Dorsalgie	dorsalgia
Dutch	Pijn, rug-	dorsalgia
Norwegian	Ryggsmerter	dorsalgia
Croatian	BOL U LEĐIMA	dorsalgia
Swedish	Ryggsmärta	dorsalgia
Chinese	背痛	dorsalgia

4.5. Diseases database relation classification

We introduce the Diseases Database relation classification (DDBRC) task to classify the relation type using head and tail concept embeddings.

Table 7
Top-3 similar terms searched from ICD-9-CM.

Top-k	sleep too much	stomachache
	$\text{CODER}_{\text{ENG}}$	
1	**hypersomnia**	**abdominal pain**
2	lack of adequate sleep	gastric ulcer
3	insomnia	gastroparesis
	$\text{CODER}_{\text{ENG}}$ $-\mathcal{S}_{ij}^{rel}$	
1	polyphagia	heartburn
2	**hypersomnia**	gastroparesis
3	insomnia	throat pain

A total of 131,741 medical relation triplets were collected from the Diseases Database[1]. A triplet contains a head concept, a tail concept, and a relationship that can be classified into 14 categories. A total of 18,419 head and tail concept pairs overlapped with those in UMLS (though the concepts may have different recorded relation types). We randomly split the data 4:1 into training and test sets.

Different embeddings are evaluated with the same classification model. We denote the embedding of the head concept as $\boldsymbol{e}_h$ and that of the tail concept as $\boldsymbol{e}_t$. The relation is predicted by:

[1] http://www.diseasesdatabase.com/.

$\hat{y} = softmax(W[\boldsymbol{e}_h, \boldsymbol{e}_t] + \boldsymbol{b})$.

We use cross-entropy as the loss function. Embeddings are tested as fixed (feature based) or trainable (fine-tuned). Further training details are listed in Appendix A.2. The classification accuracy on the test set is used as the evaluation metric.

The results of DDBRC are listed in Table 4. Contextual embeddings are powerful when fine-tuned. Whether embeddings are trainable or not, CODER always achieves the highest accuracy. Under the feature-based setting, the results of $\text{CODER}_{\text{ENG}}$ improve 5.50, 2.45, and 2.65 over the best concept, word, and contextual embeddings, respectively. CODER also displays the minimum performance difference between feature-based and fine-tuned results among all contextual embeddings, which indicates that the medical knowledge encoded by CODER yields informative machine learning features. $\text{CODER}_{\text{ALL}}$ performs better than $\text{CODER}_{\text{ENG}}$ in feature setting may be due to longer training schedule. Comparing CODER with SapBERT, $\text{CODER}_{\text{ENG}}$ improves 2.65 in feature setting, which confirms that relational knowledge in the UMLS is helpful for DDBRC.

5. Discussion

5.1. Ablation study

To explore how term-relation-term similarity impacts normalization performance, we first trained CODER without term-relation-term similarity. We further tested another TransE-like term-relation-term similarity function $S_{ij}^{rel,T} = -|e_i + r_i - e_j|$. Details of the pretraining parameters are listed in Appendix A.3.

Table 5 shows ablation results for the Cadec and PsyTar datasets. Term normalization performance is improved by considering relation information from UMLS. $S_{ij}^{rel,T}$ utilizes the relation information of UMLS, but it has an unbounded range and is inconsistent with the form of $S_{ij} = \text{cosine}(\boldsymbol{e}_i, \boldsymbol{e}_j)$; therefore, using S_{ij}^{rel} is a better choice to encode relational knowledge and improve normalization performance.

5.2. Embedding visualization analysis

We use t-SNE [58] to visualize different medical embeddings. Fig. 2 and Fig. 3 plot medical terms from the International Classification of Diseases ICD-10-CM. Terms attached to the same concept are shown in the same color in Fig. 2, and concepts are color coded by semantic type in Fig. 3. The results show that $\text{CODER}_{\text{ENG}}$ and SapBERT cluster terms automatically and group concepts according to semantic types, while BioBERT and PubMedBERT are incapable of clustering. Note that $\text{CODER}_{\text{ENG}}$ can cluster terms better and separate concepts in different categories more clearly than SapBERT, which confuses signs and symptoms, diseases and syndromes, and injuries and poisoning to some extent.

5.3. Case study of CODER

5.3.1. Cross-lingual capability

To illustrate the ability of $\text{CODER}_{\text{ALL}}$ to normalize cross-lingual terms, we search the closest English term to terms in different languages for concept **C0004604** *dorsalgia* as a case study; notably, there is no public medical normalization dataset for several languages. We searched for the most similar English terms based on the cosine distance among all ICD-9-CM concepts, and the results are shown in Table 6. Most letter-based languages are correctly normalized to the term *dorsalgia,* and the performance is not as good for character-based languages (Chinese, Japanese, and Korean). Letter-based languages benefit from sharing similar subwords, such as *dolor*. UMLS does not include Chinese terms, but CODER can normalize 背痛 because Chinese shares characters with Japanese. Unfortunately, $\text{CODER}_{\text{ALL}}$ could not handle the unique Japanese and Korean alphabets well.

5.3.2. Knowledge-injected term normalization

Table 7 is an ablation study to contrast the normalization performance when using and not using relational knowledge. We list the top similar terms for *sleep too much* and *stomachache* from ICD-9-CM. With additional knowledge from the UMLS, $\text{CODER}_{\text{ENG}}$ can normalize *sleep too much* to *hypersomnia*, and top similar terms are all related to sleep. $\text{CODER}_{\text{ENG}}$ can normalize *stomachache* to *abdominal pain*, which may imply that $\text{CODER}_{\text{ENG}}$ can understand *abdominal* is close to *stomach* and *pain* is close to *ache*. In addition, the other two top similar terms are both stomach disorders that involve stomachache. Furthermore, similar terms found by $\text{CODER}_{\text{ENG}}$ are all related to *sleep* or *stomach*, and the information apparently comes from relation rather than text, while $\text{CODER}_{\text{ENG}}$–$S_{ij}^{rel}$ finds unrelated terms. These phenomena show that relational knowledge is important for meaningful biomedical term embedding and normalization.

6. Conclusion

In this paper, we present a medical term embedding model called CODER. The KG-based contrastive learning framework learns both term-term similarity and term-relation-term similarity. Evaluations show that CODER substantially outperforms existing medical embedding methods in zero-shot term normalization and that CODER exhibits good performance in semantic similarity and relation classification. $\text{CODER}_{\text{ALL}}$ is a cross-lingual medical term embedding model and can promote the use of low-resource languages in medical NLP tasks.

7. Data availability

The data and knowledge graph used in this article are publicly available online via the following links.

UMLS: https://www.nlm.nih.gov/research/umls/licensedcontent/umlsarchives04.html#2020AA

Cadec: https://disk.yandex.com/d/GZoWm1wBxzyW_w

Psytar: https://zenodo.org/record/3236318

The mantra gsc: https://files.ifi.uzh.ch/cl/mantra/gsc/GSC-v1.1.zip

Besides, DDBRC relation triplets were collected from http://www.diseasedatabase.com/.

8. Code availability

Codes and models are available at https://github.com/GanjinZero/CODER.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Authors' contributions

All authors made substantial contributions to the conception and design; acquisition, analysis and interpretation of data; drafting the article or revising it critically for important intellectual content; and

final approval of the version to be published.

Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jbi.2021.103983.

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